

Contact Manager Sprint 1

Project Structure

```
ContactManager/
|
├── src/main/java/com/wipro/
|   ├── Contact.java
|   ├── ContactServlet.java
|   ├── ContactStorage.java
|   └── HomeServlet.java
|
├── src/main/webapp/
|   ├── style.css
|   ├── WEB-INF/
|   │   ├── views/
|   │   │   ├── index.jsp
|   │   │   └── contacts.jsp
|   └── web.xml
```

Application Flow

1. User opens Contact Manager homepage
2. User clicks Add Contact button
3. Contact form becomes visible
4. User enters contact details
5. Servlet receives form data
6. Data stored in ArrayList collection
7. Contacts displayed in table format

Key Classes

1. **Contact.java:** Stores contact details such as name, phone, and email.

```
package com.wipro;

public class Contact {

    private String name;

    private String phone;

    private String email;
```

```
public Contact() {  
    }  
  
    public Contact(String name, String phone, String email) {  
        this.name = name;  
        this.phone = phone;  
        this.email = email;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
  
    public String getPhone() {  
        return phone;  
    }  
  
    public void setPhone(String phone) {  
        this.phone = phone;  
    }  
  
    public String getEmail() {  
        return email;  
    }  
  
    public void setEmail(String email) {  
        this.email = email;  
    }  
}
```

2. ContactStorage.java: Maintains shared ArrayList for contacts.

```
package com.wipro;

import java.util.ArrayList;

import java.util.List;

public class ContactStorage {

    public static List<Contact> contacts =

        new ArrayList<>();

}
```

3. ContactServlet.java: Handles form submission and forwards data to JSP page.

```
package com.wipro;

import java.io.IOException;

import jakarta.servlet.ServletException;

import jakarta.servlet.annotation.WebServlet;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/addContact")

public class ContactServlet extends HttpServlet {

    @Override

    protected void doPost(

        HttpServletRequest request,

        HttpServletResponse response)

        throws ServletException, IOException {

        String name =

            request.getParameter("name");

        String phone =
```

```

        request.getParameter("phone");

String email =

        request.getParameter("email");

Contact contact =

        new Contact(name, phone, email);

ContactStorage.contacts.add(contact);

request.setAttribute(

        "contactList",

        ContactStorage.contacts

);

request.getRequestDispatcher(

        "/WEB-INF/views/contacts.jsp"

).forward(request, response);

}

}

```

4. **HomeServlet.java:** Loads homepage JSP from WEB-INF/views.

```

package com.wipro;
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/")
public class HomeServlet extends HttpServlet {
    @Override
    protected void doGet(
        HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        request.getRequestDispatcher(
            "/WEB-INF/views/index.jsp"
        ).forward(request, response);
    }
}

```

5. index.jsp: Displays homepage and contact form.

```
<%@ page language="java"
contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>

<!DOCTYPE html>

<html>

<head>

<meta charset="UTF-8">

<title>Contact Manager</title>

<link rel="stylesheet" href="<%=request.getContextPath()%>/style.css">

</head>

<body>

<div class="main-container">

    <div class="card">

        <div class="header">

            <h1>Contact Manager</h1>

            <p>
                Manage your contacts efficiently
            </p>

        </div>

        <button
            class="btn"
            onclick="showForm()">

            + Add Contact

        </button>

        <div
            id="formBox"
            class="form-container"
            style="display:none;">

            <form
                action="addContact"
                method="post">

                <div class="input-box">

                    <label>Full Name</label>

                    <input
                        type="text"
                        name="name"
                        placeholder="Enter Name"
                        required>
```

```

        </div>

        <div class="input-box">

            <label>Phone Number</label>

            <input
                type="text"
                name="phone"
                placeholder="Enter Phone"
                required>

        </div>

        <div class="input-box">

            <label>Email Address</label>

            <input
                type="email"
                name="email"
                placeholder="Enter Email"
                required>

        </div>

        <button
            type="submit"
            class="btn save-btn">

            Save Contact

        </button>

    </form>

</div>

</div>

<script>

function showForm() {

    document
        .getElementById("formBox")
        .style.display = "block";
}

</script>

</body>

</html>

```

6. contacts.jsp: Displays saved contacts in table format.

```

<%@ page language="java"
contentType="text/html; charset=UTF-8"

```

```

pageEncoding="UTF-8"%>

<%@ page import="java.util.List" %>
<%@ page import="com.wipro.Contact" %>

<!DOCTYPE html>

<html>

<head>

<meta charset="UTF-8">

<title>Saved Contacts</title>

<link rel="stylesheet"
href="<%= request.getContextPath() %>/style.css">

</head>

<body>

<div class="main-container">

    <div class="table-card">

        <div class="header">

            <h1>Saved Contacts</h1>

            <p>
                Your contact information is listed below
            </p>

        </div>

        <div class="table-wrapper">

            <table>

                <tr>

                    <th>Name</th>
                    <th>Phone</th>
                    <th>Email</th>

                </tr>

                <%

                    List<Contact> contacts =
                    (List<Contact>)
                    request.getAttribute("contactList");

                    if (contacts != null) {

                        for (Contact c : contacts) {

                            <%>

                            <tr>

                                <td><%= c.getName() %></td>

```

```

        <td><%= c.getPhone() %></td>

        <td><%= c.getEmail() %></td>

    </tr>

    <%

        }

    %>

</table>

</div>

<a href=".." class="btn add-btn">

+ Add More Contacts

</a>

</div>

</div>

</body>
</html>

```

7. style.css:

```

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}

body {
    min-height: 100vh;
    background: linear-gradient(
        135deg,
        #1d2671,
        #c33764
    );
}

.main-container {
    width: 100%;
    min-height: 100vh;
    display: flex;

```



```
    justify-content: center;  
    align-items: center;  
    padding: 20px;  
}
```

```
.card,  
.table-card {  
    width: 100%;  
    max-width: 750px;  
    background: white;  
    border-radius: 18px;  
    padding: 35px;  
    box-shadow: 0 12px 35px rgba(0,0,0,0.3);  
}
```

```
.header {  
    text-align: center;  
    margin-bottom: 25px;  
}
```

```
.header h1 {  
    color: #1d2671;  
    font-size: 36px;  
    margin-bottom: 10px;  
}
```

```
.header p {  
    color: #666;  
    font-size: 15px;  
}
```

```
.form-container {  
    margin-top: 25px;  
}
```

```
form {  
    display: flex;  
    flex-direction: column;  
    gap: 20px;  
}
```

```
.input-box {  
    display: flex;  
    flex-direction: column;  
    gap: 8px;  
}
```

```
label {
```

```
font-weight: bold;  
color: #333;  
}
```

```
input {  
padding: 14px;  
border: 1px solid #ccc;  
border-radius: 10px;  
font-size: 15px;  
transition: 0.3s;  
}
```

```
input:focus {  
outline: none;  
border-color: #c33764;  
box-shadow: 0 0 8px rgba(195,55,100,0.4);  
}
```

```
.btn {  
background: #1d2671;  
color: white;  
border: none;  
padding: 14px;  
border-radius: 10px;  
cursor: pointer;  
text-decoration: none;  
text-align: center;  
font-size: 15px;  
font-weight: bold;  
transition: 0.3s;  
}
```

```
.btn:hover {  
background: #c33764;  
transform: translateY(-2px);  
}
```

```
.save-btn,  
.add-btn {  
margin-top: 10px;  
}
```

```
.table-wrapper {  
overflow-x: auto;  
}
```

```
table {  
width: 100%;
```

```

border-collapse: collapse;
margin-top: 20px;
}

table th {
background: #1d2671;
color: white;
padding: 14px;
border: 1px solid #ddd;
}

table td {
padding: 14px;
border: 1px solid #ddd;
text-align: center;
}

table tr:nth-child(even) {
background: #f8f8f8;
}

table tr:hover {
background: #f1f1f1;
}

@media(max-width: 600px) {
.card,
.table-card {
padding: 25px;
}
.header h1 {
font-size: 28px;
}

table {
font-size: 13px;
}

input,
.btn {
font-size: 14px;
}
}

```

8. web.xml:

```

<?xml version="1.0" encoding="UTF-8"?>
<web-app

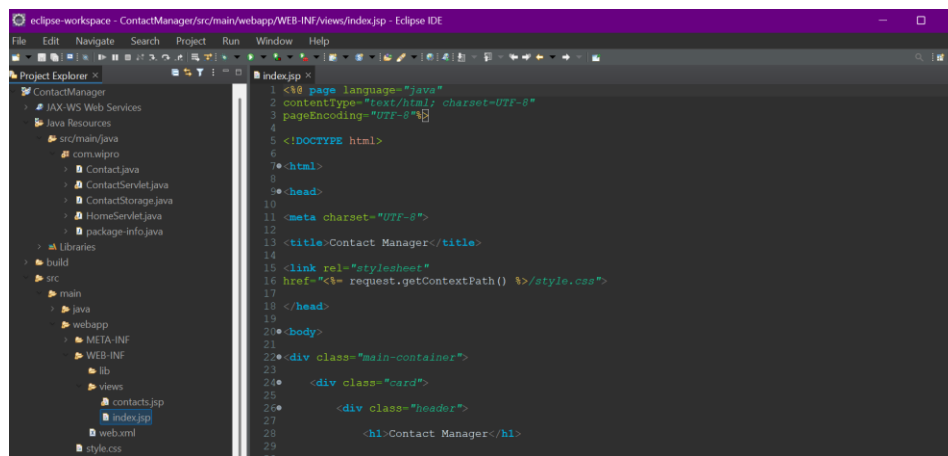
```

```

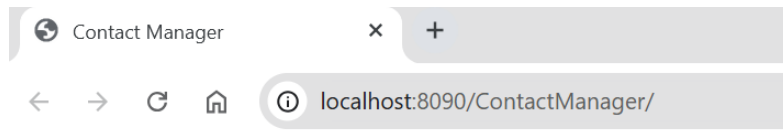
xmlns="https://jakarta.ee/xml/ns/jakartaee"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
https://jakarta.ee/xml/ns/jakartaee
https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
version="6.0">
<welcome-file-list>
  <welcome-file>index.jsp</welcome-file>
</welcome-file-list>
</web-app>

```

Output:



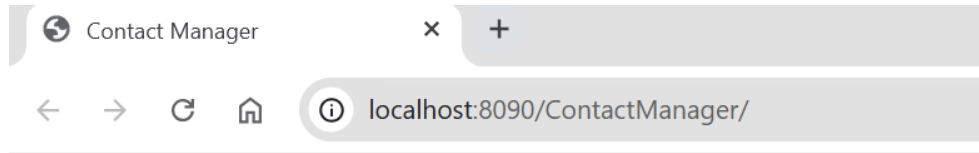
The screenshot shows the Eclipse IDE interface. On the left, the Project Explorer displays the project structure for 'ContactManager'. The main editor window shows the content of 'index.jsp', which is an XML file for a web application. The code includes XML declarations for namespaces, a DOCTYPE declaration, and a header section with a title 'Contact Manager' and a link to a stylesheet. The body of the page contains a main container and a header section with a title 'Contact Manager'.



Contact Manager

Manage your contacts efficiently

[+ Add Contact](#)



Contact Manager

Manage your contacts efficiently

+ Add Contact

Full Name

Aishu

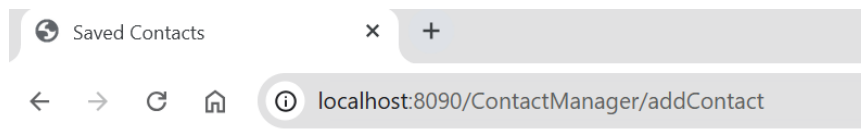
Phone Number

9988776655

Email Address

aishu@gmail.com

Save Contact



Saved Contacts

Your contact information is listed below

Name	Phone	Email
Aishu	9988776655	aishu@gmail.com

[+ Add More Contacts](#)